

Ordinary Least Squares - brief recap

Let (x_i, y_i) be data points in $\mathbb{R}^d \times \mathbb{R}$ and consider the family of linear predictors $\mathcal{F} = \{f_\theta : x \in \mathbb{R}^d \mapsto x^\top \theta = \langle x, \theta \rangle \mid \theta \in \Theta\}$, with the parameter space being $\Theta = \mathbb{R}^d$. Define the design matrix and response vector by

$$X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{pmatrix} \in \mathbb{R}^{n \times d}, \quad y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n,$$

The ordinary least-squares objective is

$$\hat{R}_n(\theta) = \frac{1}{2n} \|y - X\theta\|_2^2.$$

- Its gradient and Hessian are

$$\nabla \hat{R}_n(\theta) = \frac{1}{n} X^\top (X\theta - y), \quad \nabla^2 \hat{R}_n(\theta) = \frac{1}{n} X^\top X.$$

- Every OLS solution satisfies the **normal equations**

$$X^\top X \hat{\theta} = X^\top y.$$

- If X has full column rank, then $X^\top X$ is invertible and the OLS solution is unique:

$$\hat{\theta} = (X^\top X)^{-1} X^\top y.$$

- If X does not have full column rank, the solution need not be unique. The Moore–Penrose pseudoinverse gives the unique solution of smallest Euclidean norm:

$$\hat{\theta}^\dagger = X^+ y.$$

For Exercises 1 and 2, observations follow the same model as in TD3 but without outlier noise:

$$y_i = w_0 x_i + b_0 + \sigma \varepsilon_i$$

where the ε_i are i.i.d. $\mathcal{N}(0, 1)$. Define

$$S_1 = \frac{1}{n} \sum_{i=1}^n x_i, \quad S_2 = \frac{1}{n} \sum_{i=1}^n x_i^2, \quad T_1 = \frac{1}{n} \sum_{i=1}^n y_i, \quad T_2 = \frac{1}{n} \sum_{i=1}^n x_i y_i, \quad E_1 = \frac{1}{n} \sum_{i=1}^n \varepsilon_i, \quad E_2 = \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i.$$

Exercise 1. The through-origin model as OLS

Consider the linear model $f_w(x) = wx$ with the empirical risk

$$\hat{R}_n(w) = \frac{1}{2n} \sum_{i=1}^n (y_i - wx_i)^2$$

and assume $S_2 > 0$.

1. Show that the empirical quadratic risk from TD3 can be written as $\hat{R}_n(w) = \frac{1}{2n} \|y - Xw\|_2^2$ and give the dimensions of y , X , and w .
2. Compute $X^\top X$ and $X^\top y$. Apply the OLS formula and recover the expression for $\hat{w}$ obtained in TD3.

Exercise 2. The affine model as two-dimensional OLS

Consider this time the affine model $f_{w,b}(x) = wx + b$ and assume the x_i are not all equal.

1. Show that for a given data matrix $Z(x)$ that depends on the inputs x_i and a well chosen vector θ , the empirical risk can be written as $\hat{R}_n(\theta) = \frac{1}{2n} \|y - Z(x)\theta\|_2^2$. What are the dimensions of $Z(x)$ and θ ?
2. Compute $\frac{1}{n} Z(x)^\top Z(x)$ and $\frac{1}{n} Z(x)^\top y$ in terms of S_1, S_2, T_1, T_2 .
3. Show that $Z(x)$ has full column rank if and only if $S_2 - S_1^2 > 0$. Interpret this condition in terms of the empirical variance of the inputs.

4. Apply the OLS formula and recover the TD3 expressions for $(\hat{w}, \hat{b})$.

Exercise 3. Rank, pseudoinverse, and singular value decomposition

Let $X \in \mathbb{R}^{n \times d}$ be arbitrary; it is not assumed to have full column rank. Consider

$$\hat{R}_n(\theta) = \frac{1}{2n} \|y - X\theta\|_2^2.$$

1. For $\delta \in \mathbb{R}^d$ expand $\hat{R}_n(\theta + \delta)$ as a quadratic function of δ and show that we have

$$\hat{R}_n(\theta + \delta) = \hat{R}_n(\theta) + \nabla \hat{R}_n(\theta)^\top \delta + \frac{1}{2} \delta^\top \nabla^2 \hat{R}_n(\theta) \delta$$

2. Show that the Hessian is positive semidefinite. When is it positive definite?
 3. Show that every minimizer satisfies the normal equations.
 4. Let $r = \text{rank}(X)$ and consider a singular value decomposition

$$X = U \Sigma V^\top = \sum_{i=1}^r \sigma_i u_i v_i^\top,$$

where $U \in \mathbb{R}^{n \times n}$ and $V \in \mathbb{R}^{d \times d}$ are orthogonal, and the nonzero singular values satisfy $\sigma_1 \geq \dots \geq \sigma_r > 0$. Define Σ^+ by replacing each nonzero σ_j by $1/\sigma_j$ and transposing the rectangular matrix. Recall that

$$X^+ = V \Sigma^+ U^\top = \sum_{i=1}^r \frac{1}{\sigma_i} v_i u_i^\top.$$

Show that $\text{Im}(X) = \text{Vect}(u_1, \dots, u_r)$ and that $\ker(X) = \text{Vect}(v_{r+1}, \dots, v_d)$. What is $\ker(X)^\perp$?

5. Show that XX^+ is the orthogonal projector onto $\text{Im}(X)$ and that X^+X is the orthogonal projector onto $\text{Ker}(X)^\perp$.
 6. Prove that $\theta^\dagger = X^+y$ is an OLS solution. Show that every OLS solution can be written as

$$\theta = X^+y + (I_d - X^+X)z, \quad z \in \mathbb{R}^d.$$

7. Compute the Euclidean norm of a general OLS solution and deduce that $\theta^\dagger$ is the unique OLS solution with minimum Euclidean norm.